ENGINEERING PHYSICS LAB

(23AS6101P) | Common to All Branches of Engineering

B.Tech - Artificial Intelligence & Machine Learning | JNTUA R23 Regulations

I Year I Semester | Easy handwritten-style lab notes with diagrams

L	T	P	C
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Course Objectives

To study the concepts of **optical phenomena** like interference and diffraction, recognise the importance of the **energy gap** in conductivity and the **Hall effect** in semiconductors, and study the parameters and applications of **dielectric and magnetic materials** by conducting experiments.

Course Outcomes

CO	The students will be able to
CO1	Operate optical instruments like **travelling microscope** and **spectrometer**.
CO2	Estimate the **wavelengths** of different colours using a **diffraction grating**.
CO3	Plot the **intensity of the magnetic field** of a circular coil carrying current with distance.
CO4	Evaluate **dielectric constant** and **magnetic susceptibility** of dielectric and magnetic materials.
CO5	Calculate the **band gap** of a given semiconductor.
CO6	Identify the **type of semiconductor** using the Hall effect.

* **Study tip:**
Any **TEN** experiments are to be conducted. Out of these, any **TWO** may be done in **virtual mode** (vlab.co.in or PhET simulations). For every experiment learn: **Aim -> Formula -> Diagram -> Procedure -> Table -> Result**.

List of Experiments (with course outcome)

No.	Experiment	CO
1	Radius of curvature of a plano-convex lens by **Newton's rings**	CO1
2	Wavelengths of spectral lines of **mercury** using a diffraction grating (normal incidence)	CO1, CO2
3	Verification of **Brewster's law**	-
4	**Dielectric constant** by charging and discharging method	CO4
5	**B-H curve** of a magnetic material	CO4
6	Wavelength of **laser light** using a diffraction grating	CO2
7	**Planck's constant** using the photoelectric effect	-
8	Resistivity of semiconductors by the **four probe method**	CO5
9	**Energy gap** of a semiconductor using a p-n junction diode	CO5
10	Magnetic field along the axis of a circular coil - **Stewart & Gee**	CO3
11	**Hall voltage and Hall coefficient** of a semiconductor	CO6
12	**Temperature coefficient** of a thermistor	-
13	Acceleration due to gravity and radius of gyration - **compound pendulum**	-
14	**Magnetic susceptibility** by Kundt's tube method	CO4
15	**Rigidity modulus** by torsional pendulum	-
16	**Sonometer**: laws of a stretched string	-
17	**Young's modulus** by non-uniform bending	-
18	Frequency of a tuning fork by **Melde's experiment**	-

General Lab Instructions

- Come **prepared**: read the aim, formula and procedure before the class.
- Check **least count** and **zero error** of every instrument first.
- Take **at least 3 readings**; write them in the table in ink, never overwrite.
- Draw **neat graphs** with a sharp pencil, labelled axes, units and scale.
- Switch off the supply after use; handle **glass, lasers and heaters** carefully.

References and web resources

- A Textbook of Practical Physics - **S. Balasubramanian, M.N. Srinivasan**, S. Chand Publishers, 2017.
- Virtual labs: **www.vlab.co.in**
- PhET simulations: **phet.colorado.edu** (physics, html prototypes).

Basics : Instruments, Errors and Graphs

Least count of common instruments

Instrument	Least count (LC)	Used in
Vernier calliper	LC = 1 MSD - 1 VSD = **value of 1 MSD / total VSD**	Dimensions of scale, disc
Screw gauge	LC = **pitch / number of head divisions** (e.g. 1 mm / 100 = 0.01 mm)	Wire radius r
Travelling microscope	LC = **1 MSD / number of VSD** (e.g. 0.05 cm / 50 = 0.001 cm)	Newton's rings, Kundt's tube
Spectrometer	LC = **1 main scale division / number of vernier divisions** (typically 30 seconds / 1 minute)	Grating, mercury lines
Stop clock	Usually **0.1 s** (digital: 0.01 s)	Pendulums

Reading = MSR + (VSR x LC)

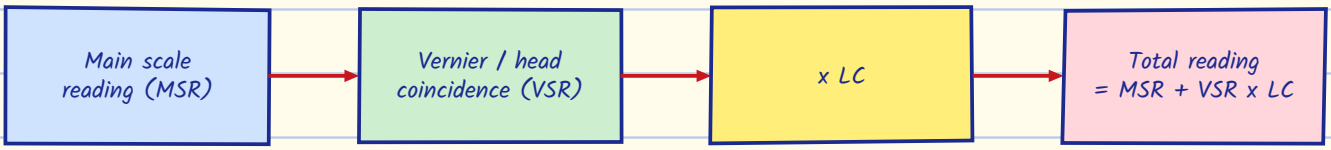


Fig 0.1 : How to take a reading

Errors and graphs

- **Mean** = sum of readings / number of readings.
- **Percentage error** = (experimental - standard) / standard x 100.
- **Zero error** is subtracted: corrected reading = observed - zero error.
- **Backlash error** - rotate screws in one direction only while taking readings.
- Graph: choose a **large scale**, mark points with a dot in a circle, draw a **best-fit straight line**, find slope using a **big triangle**.

> * **Study tip:**
> Slope = (y2 - y1) / (x2 - x1) taken from **two points far apart on the line**, not from data points.

Experiment 1 : Newton's Rings

Aim: To determine the **radius of curvature R** of a plano-convex lens by forming **Newton's rings** with sodium light.

Apparatus: Plano-convex lens, plane glass plate, glass plate at 45 deg, sodium vapour lamp (lambda = 589.3 nm), travelling microscope, lens holder, magnifying glass.

Principle / Theory

- A thin **air film** of varying thickness is formed between the curved surface of the lens and the glass plate.
- Light reflected from the top and bottom of the film **interferes**, giving **circular dark and bright rings**.
- The centre is **dark** because reflection from the denser glass plate adds a path change of lambda/2.
- For the nth dark ring the diameter satisfies **D_n^2 = 4 n lambda R**, so D^2 vs n is a straight line.

> * **Formula:**
> R = [D_(n+p)^2 - D_n^2] / (4 p lambda)
> Graph: slope of D^2 - n line = 4 lambda R, so R = slope / (4 lambda)

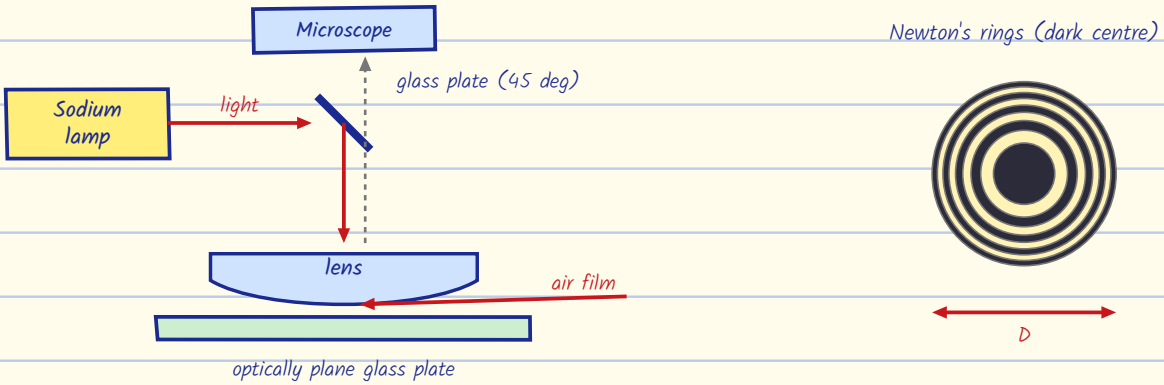


Fig 1.1 : Newton's rings set-up and ring pattern

Procedure

1. Clean the lens and plate; place the lens (curved face down) on the plate under the microscope.
2. Adjust the 45 deg glass plate so sodium light falls **vertically** on the lens.
3. Focus the microscope until sharp rings with a **dark centre** are seen; find the LC.

4. Move the cross-wire to the **left side** beyond the 16th ring, then bring it back and note readings for rings 16, 14, 12, 10, 8, 6 (move in **one direction**).
5. Cross the centre and note readings on the **right side** for the same rings.
6. Find D = (left - right) and D^2; take p = 6 (e.g. 16th and 10th) and calculate R.

Observations

Ring no. n	Left reading (cm)	Right reading (cm)	D (cm)	D^2 (cm^2)
16				
14				
12				
10				
8				
6				

Calculations and graph

- Take D^2 of two rings **p apart** and apply the formula for each pair; find the **mean R**.
- Plot **D^2 (y-axis) against n (x-axis)**: a straight line through the origin; use its slope in R = slope/(4 lambda).

Result: Radius of curvature of the lens R = ... cm (graph) and ..... cm (calculation). Typical value: about 100 cm.

Precautions

- Do not touch the lens surfaces; keep them **dust free**.
- Avoid **backlash**: move the microscope in one direction only.
- Count rings **from the centre** carefully; measure diameters, not radii.

> * **Viva voce:**
> Q1. Why is the centre dark? A: Phase change of pi on reflection at the glass plate gives destructive interference.
> Q2. What happens with white light? A: Coloured rings with a dark centre.
> Q3. Why do rings get closer outwards? A: D is proportional to sqrt(n), so spacing decreases.

Experiment 2 : Mercury Spectrum using a Diffraction Grating

Aim: To determine the **wavelengths of the spectral lines** of mercury using a diffraction grating in the **normal incidence** configuration.

Apparatus: Spectrometer, plane diffraction grating (about 15000 lines per inch), mercury vapour lamp, spirit level, reading lens.

Principle / Theory

- A grating splits light into **different directions** for each wavelength (diffraction + interference).
- In **normal incidence** the beam falls perpendicular to the grating plane.
- Grating equation: $(a + b)\sin(\theta) = n\lambda$, where $(a + b)$ = **grating element** = $1/N$.
- Different colours of mercury (violet, blue, green, yellow) appear at different angles.

> * **Formula:**
> $\lambda = \sin(\theta) / (n N)$
> N = number of lines per metre = $1/(a+b)$; 1 inch = 2.54 cm
> 15000 lines per inch = 5906 lines per cm

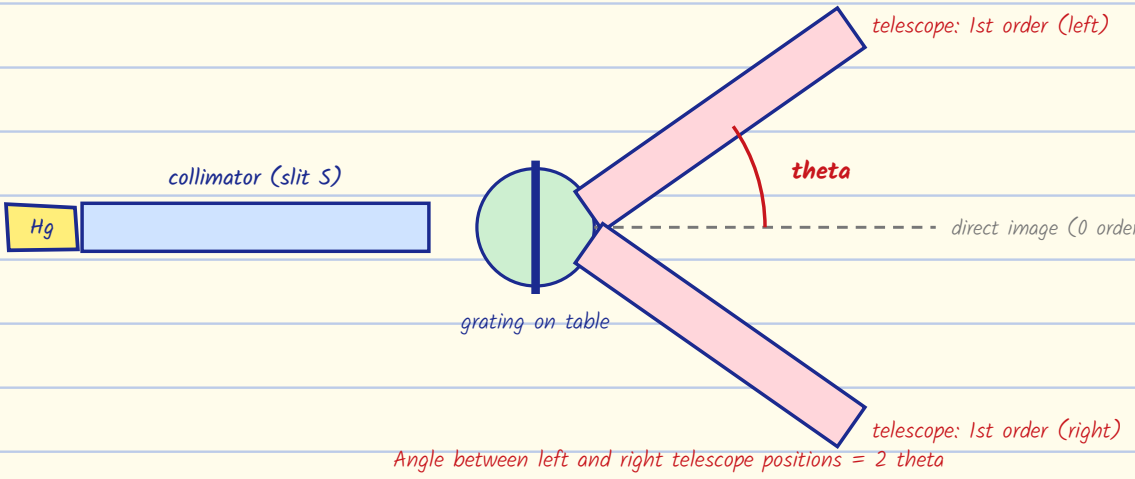


Fig 2.1 : Spectrometer arrangement for normal incidence

Procedure

1. Level the spectrometer and focus the **telescope for infinity** and the **slit** with a narrow, sharp image.
2. Take the **direct reading** of the slit image (vernier I and II).
3. Rotate the telescope exactly **90 deg** and fix it; turn the table until the slit image reflected from the grating surface is on the cross wire; fix the table.

4. Rotate the table by **45 deg** (vernier table); now the grating is at **normal incidence**; fix it.
5. Release the telescope; note readings for each coloured line in the **first order on left and right**.
6. 2 theta = difference of left and right readings (mean of both verniers). Calculate lambda for each colour.

Observations

Colour	Left (V1 / V2)	Right (V1 / V2)	2 theta	theta	lambda (nm)
Violet					
Blue					
Green					
Yellow I					
Yellow II					

Calculations and graph

- Use **both verniers** and take the mean of 2 theta; theta = (2 theta) / 2.
- Compare with standard mercury lines: **violet 404.7, blue 435.8, green 546.1, yellow 577.0 and 579.1 nm**.

Result: Wavelengths measured: violet = ... nm, blue = ... nm, green = ... nm, yellow = ... nm.

Precautions

- Spectrometer must be **levelled**; slit should be **narrow**.
- Do not touch the **grating surface**.
- Take readings with **both verniers** to remove centring error.

> * **Viva voce:**
> Q1. Why use a grating? A: It gives bright, widely separated spectra.
> Q2. What is the grating element? A: Distance between centres of two consecutive slits, 1/N.
> Q3. Why two verniers? A: To cancel errors from the axis not being at the centre.

Experiment 3 : Brewster's Law

Aim: To **verify Brewster's law** and find the **polarising angle** and refractive index of a glass plate.

Apparatus: Laser (or monochromatic source) with polariser, glass plate on a rotating table with angular scale, photo-detector / lux meter, analyser, screen.

Principle / Theory

- When unpolarised light reflects from a surface, the reflected light is **partly polarised**.
- At the **polarising (Brewster) angle iB**, the reflected and refracted rays are **perpendicular** and the reflected light is completely polarised.
- Light polarised **parallel to the plane of incidence** shows **zero reflection** at iB.
- Since iB + r = 90 deg, Snell's law gives **mu = tan iB**.

> * **Formula:**
> mu = tan(iB) iB + r = 90 deg
> For glass mu = 1.5, so iB = about 56.3 deg

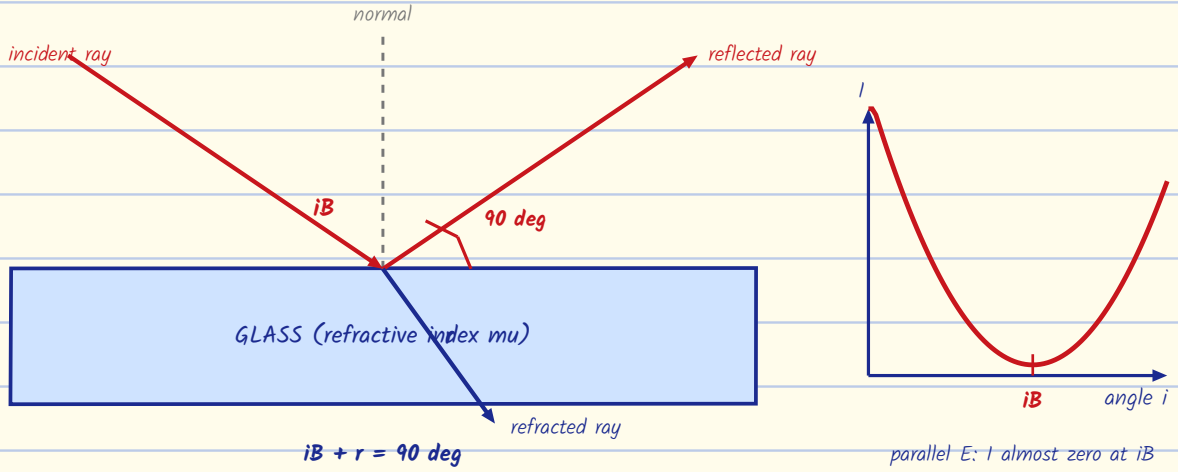


Fig 3.1 : Brewster angle: reflected and refracted rays at 90 deg

Procedure

1. Place the laser and set the polariser so the light is polarised **in the plane of incidence**.
2. Mount the glass plate on the rotating table with its surface at the centre of the scale.
3. Vary the **angle of incidence i** from 40 deg to 70 deg in steps of 2 deg; each time place the detector on the **reflected beam**.
4. Note the **reflected intensity** for every angle; the angle at which intensity is **minimum (nearly zero)** is iB.
5. Plot a graph of intensity against i and find iB; calculate mu = tan iB.

Observations

Angle of incidence i (deg)	Reflected intensity I (micro-A)	Remarks
44		

Angle of incidence i (deg)	Reflected intensity I (micro-A)	Remarks
48		
52		
54		
56		
58		
60		
64		

Calculations and graph

- From the graph read **iB** at the minimum of intensity.
- Calculate mu = tan(iB) and compare with the standard value for glass (about **1.5**).

Result: Polarising angle iB = ... deg, refractive index of glass mu = tan iB = ...

Precautions

- Never look **directly into the laser beam**.
- Keep the plate surface exactly at the **centre of the rotation axis**.
- Measure in a **dark room** to avoid stray light.

> * **Viva voce:**
> Q1. State Brewster's law. A: tan iB = mu.
> Q2. Which light is not reflected at iB? A: Light polarised in the plane of incidence.
> Q3. Angle between reflected and refracted ray at iB? A: 90 deg.

Experiment 4 : Dielectric Constant by Charging and Discharging

Aim: To determine the **dielectric constant** of a material by the **charging and discharging** of a capacitor through a resistor.

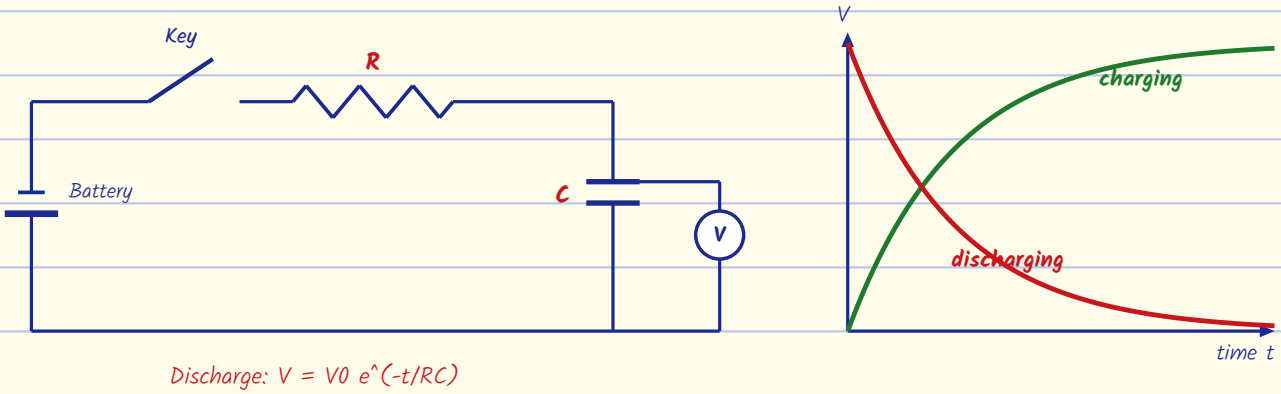
Apparatus: Parallel plate capacitor with dielectric slab, DC supply (6-10 V), high resistance R (about 1 Mohm), high-impedance digital voltmeter, two-way key, stop clock.

Principle / Theory

- A capacitor C charged through R follows **V = V0 (1 - e^(-t/RC))** and discharges as **V = V0 e^(-t/RC)**.
- Taking logs: **ln V = ln V0 - t/(RC)**; a plot of ln V vs t is a straight line with slope **-1/(RC)**.

- Insert the dielectric: capacitance rises to **K times** the air value (K = dielectric constant).
- C = K epsilon0 A / d.

> * **Formula:**
> C = 1 / (R x |slope|) K = C (with dielectric) / C0 (air)
> Or K = C d / (epsilon0 A), epsilon0 = 8.85 x 10^-12 F/m



Discharge: V = V0 e^(-t/RC)

Fig 4.1 : RC circuit and charging / discharging curves

Procedure

1. Connect the circuit as shown; first with **air** between the plates.
2. Charge the capacitor to V0 (say 6 V), then switch the key to **discharge** through R.
3. Start the stop clock and note the voltage **every 10 s** until it falls to about 1 V.
4. Plot **ln V against t**; find the slope and calculate C0 = 1/(R |slope|).
5. Repeat with the **dielectric slab** inserted to get C.
6. Calculate K = C / C0 and compare with the standard value.

Observations

Time t (s)	V (air) (V)	ln V	V (dielectric) (V)	ln V
0				
10				
20				
30				
40				
50				
60				

Calculations and graph

- Slope of ln V vs t (air) = ... s^-1 -> C0 = ... F.
- Slope with dielectric = ... s^-1 -> C = ... F; K = C / C0.

Result: Dielectric constant of the given material K = ...

Precautions

- Use a **high-impedance voltmeter**, otherwise it discharges the capacitor.
- Keep the plates **parallel and clean**.
- Handle the **polarity** of an electrolytic capacitor correctly.

> * **Viva voce:**
> Q1. What is the time constant? A: tau = RC, time for V to fall to 37 %.
> Q2. Why does C rise with a dielectric? A: Polarisation of the dielectric reduces the field, so more charge is stored.
> Q3. Unit of epsilon0? A: F/m.

Experiment 5 : B-H Curve of a Magnetic Material

Aim: To study the **variation of B with H** by magnetising a ferromagnetic material (B-H / hysteresis curve) and to find **retentivity, coercivity and hysteresis loss**.

Apparatus: B-H curve tracer (or toroid specimen with primary and secondary windings), AC supply, rheostat, resistors R1, R2, capacitor C, CRO, graph paper.

Principle / Theory

- When a ferromagnetic material is magnetised and demagnetised, **B lags behind H** - this is **hysteresis**.
- **Retentivity (Br)**: B left when H = 0. **Coercivity (Hc)**: reverse H needed to make B = 0.
- **Area of the loop** = energy lost per unit volume per cycle (as heat).
- Soft iron has a **narrow loop** (low loss); steel has a **wide loop** (permanent magnets).

> * **Formula:**
> H = N1 Vx / (L R1) (A/m) B = R2 C Vy / (N2 A) (T)
> Energy loss per cycle per unit volume = area of loop (J/m^3)

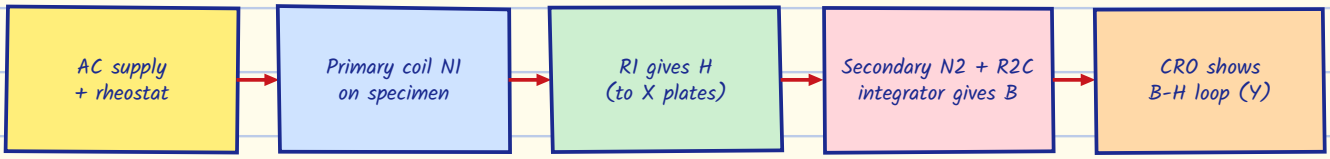


Fig 5.1 : Block diagram of the B-H tracer

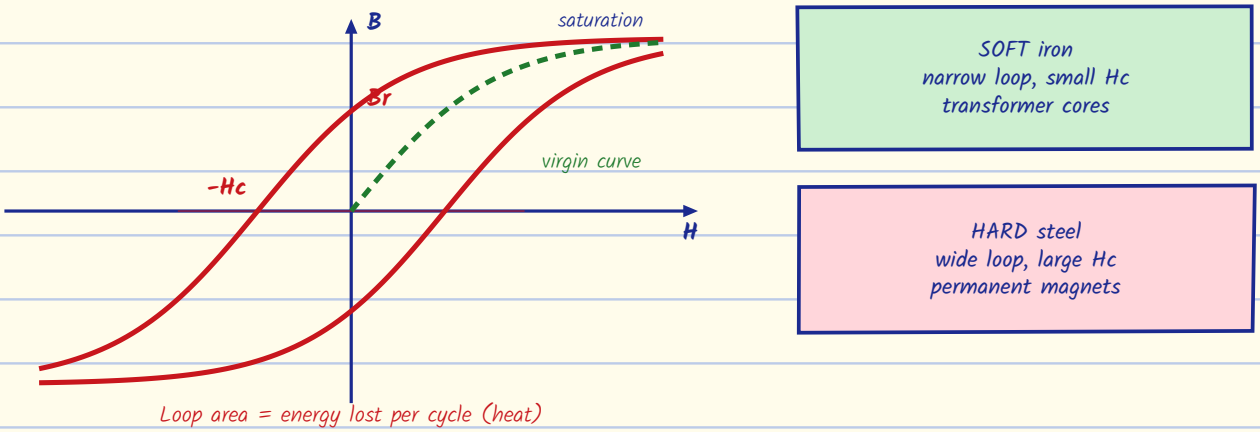


Fig 5.2 : B-H (hysteresis) loop

Procedure

1. Connect the specimen coil and the CRO as per the block diagram; keep the CRO in **X-Y mode**.
2. Increase the current slowly till the loop just **saturates** (trace becomes flat at the ends).
3. Trace the loop on tracing paper and **calibrate** the X and Y scales (cm per unit).
4. Read **Br** (cut on the B axis) and **Hc** (cut on the H axis).
5. Find the **area of the loop** by counting squares and multiply by the scale factors.
6. Repeat for a **second material** (soft iron and steel) for comparison.

Observations

Material	Br (T)	Hc (A/m)	Loop area (cm^2)	Loss (J/m^3 per cycle)
Soft iron				
Steel				

Calculations and graph

- Area of loop (J/m^3) = squares x (H per cm) x (B per cm).
- Plot **B vs H** and mark Br, Hc and saturation clearly.

Result: Retentivity Br = ... T, coercivity Hc = ... A/m, hysteresis loss = ... J/m^3 per cycle.

Precautions

- Do not **over-saturate**; the supply must be steady.
- Use **short leads** to avoid noise on the CRO.
- Switch off the current **slowly** to avoid overshoot.

> * **Viva voce:**
> Q1. What is hysteresis? A: Lag of B behind H.
> Q2. Which material for transformers? A: Soft iron (small loop).
> Q3. What does loop area represent? A: Energy loss per unit volume per cycle.

Experiment 6 : Laser Wavelength using a Grating

Aim: To determine the **wavelength of laser light** using a diffraction grating.

Apparatus: He-Ne or diode laser, diffraction grating (N lines per mm), screen, metre scale, stand.

Principle / Theory

- A grating produces **bright spots (orders n = 0, +-1, +-2)** on a screen.
- Condition for maxima: **d sin(theta) = n lambda**, with d = 1/N.
- From the geometry, tan(theta) = x_n / D.

> * **Formula:**
> lambda = d sin(theta) / n d = 1/N
> tan(theta) = x_n / D He-Ne laser: 632.8 nm

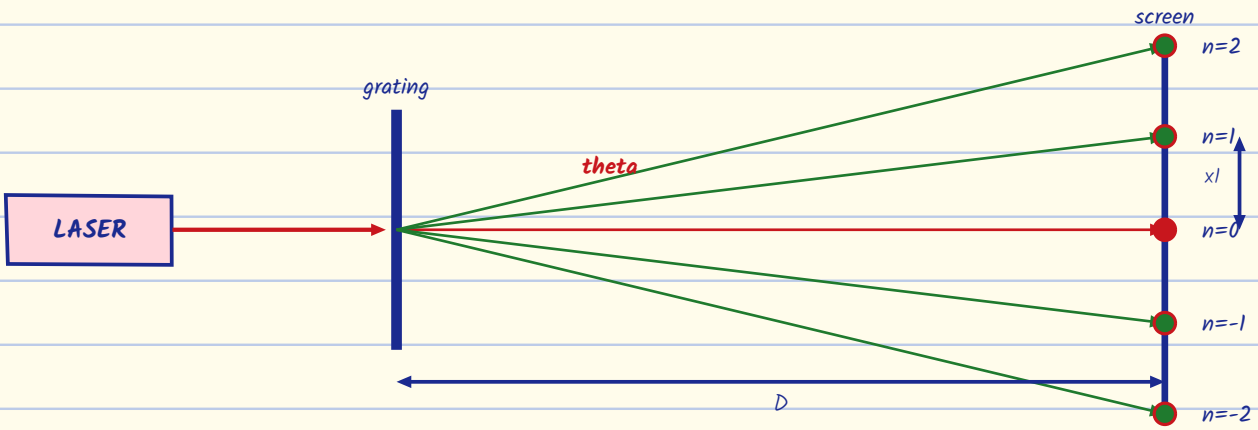


Fig 6.1 : Laser and grating arrangement

Procedure

1. Fix the laser and grating so the beam falls **normally** on the grating.

2. Place the screen at a distance **D (about 100 cm)** from the grating.
3. Mark the **central spot** and the first and second order spots on both sides.
4. Measure the distance **x_n** from the central spot to each order spot.
5. Repeat for **different D** (e.g. 80, 100, 120 cm).
6. Calculate theta and lambda for each reading and take the mean.

Observations

D (cm)	Order n	x_n (cm)	tan(theta) = x/D	sin(theta)	lambda (nm)
	1				
	2				
	1				
	2				

Calculations and graph

- theta = tan^-1 (x_n / D); then lambda = d sin(theta) / n.
- Mean lambda = ... nm; compare with standard 632.8 nm (He-Ne) and find the **% error**.

Result: Wavelength of the laser light lambda = ... nm.

Precautions

- **Never look into the laser**; avoid reflecting it into eyes.
- The grating must be **perpendicular** to the beam.
- Measure x from **spot centres**; use a screen in a dim room.

> * **Viva voce:**
> Q1. What is a grating? A: A plate with many equally spaced fine slits.
> Q2. Why is laser used? A: It is monochromatic, bright and highly directional.
> Q3. Maximum order possible? A: n < d / lambda.

Experiment 7 : Planck's Constant by Photoelectric Effect

Aim: To estimate **Planck's constant h** using the photoelectric effect.

Apparatus: Photocell, light source, set of colour filters (known wavelengths), DC variable supply, voltmeter, micro-ammeter, connecting wires.

Principle / Theory

- Light of frequency above the **threshold** knocks out electrons from a metal surface.
- Einstein: **e V0 = h ν − phi**; V0 = **stopping potential** (retarding voltage that makes photocurrent zero).
- V0 vs ν is a **straight line** of slope h/e; its intercept on the ν axis gives the **threshold frequency ν0**.

> * **Formula:**
> h = e × slope = e (V2 − V1) / (ν2 − ν1) ν = c / lambda
> Work function phi = h ν0 e = 1.6 × 10^-19 C, c = 3 × 10^8 m/s
> Standard h = 6.626 × 10^-34 J s

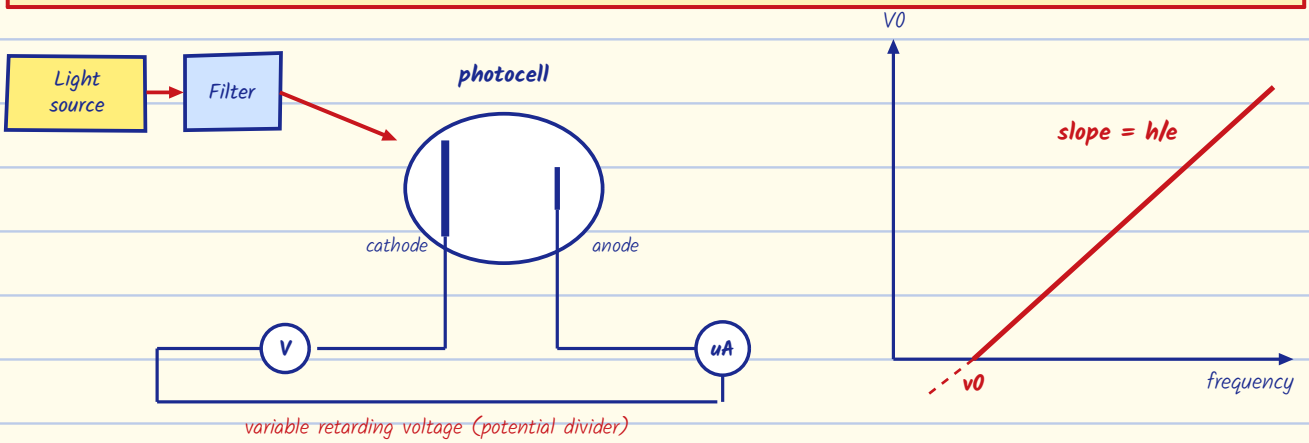


Fig 7.1 : Photoelectric effect circuit and V0 - frequency graph

Procedure

1. Connect the photocell with a **retarding (reverse) voltage** as shown.
2. Place a colour filter in front of the cell; keep light **intensity constant**.
3. Increase the retarding voltage until the **photocurrent becomes zero**; note this **V0**.
4. Repeat for every filter of different wavelength.
5. Calculate ν = c / lambda for each filter.
6. Plot **V0 vs ν**; find the slope and compute h = e × slope.

Observations

Filter (colour)	lambda (nm)	ν (× 10^14 Hz)	Stopping potential V0 (V)
Red			
Yellow			
Green			
Blue			
Violet			

Calculations and graph

- Choose two **far apart points** on the line to find slope = (V2 - V1)/(v2 - v1).
- Extrapolate to V0 = 0 to read the **threshold frequency v0**; then phi = h v0 (in eV = V0 intercept).

Result: Planck's constant h = ... J s (standard 6.626 x 10^-34 J s); work function phi = ... eV.

Precautions

- Keep the **intensity and distance constant** for all filters.
- Avoid **stray light**; cover the set-up.
- Do not exceed the **rated voltage** of the photocell.

* **Viva voce:**
Q1. What is stopping potential? A: Reverse voltage that stops the fastest photoelectrons.
Q2. Does V0 depend on intensity? A: No, only on frequency.
Q3. What is the threshold frequency? A: Minimum frequency that can emit electrons.

Experiment 8 : Resistivity by the Four Probe Method

Aim: To determine the **resistivity of a semiconductor** by the **four probe method**.

Apparatus: Four probe set-up (probes equally spaced s), semiconductor (Ge) sample, constant current source, digital milli-voltmeter, oven with temperature control (optional).

Principle / Theory

- Four collinear **equally spaced probes** touch the sample: current I flows through the **outer** two and voltage V is measured across the **inner** two.
- This removes the **contact resistance** error of the two-probe method.
- For a thin slab (thickness w less than s/2) the resistivity is given by the formula below.
- Optional: in the intrinsic region **rho is proportional to exp(Eg/2kT)** - gives the band gap.

* **Formula:**
rho = (V/I) x (pi w / ln 2) = 4.532 w (V/I) (thin slab)
For a thick sample: rho0 = 2 pi s (V/I)
Eg = 2 k x slope of (ln rho vs 1/T) ; k = 8.617 x 10^-5 eV/K

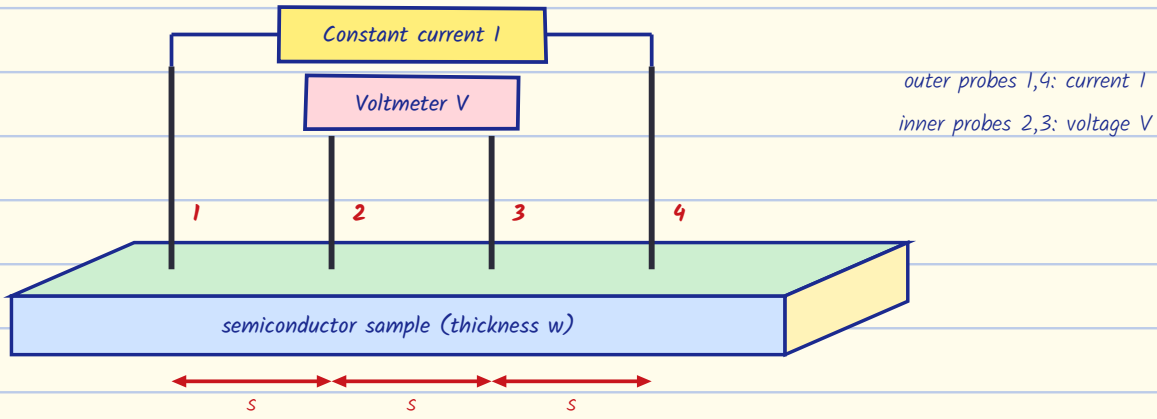


Fig 8.1 : Four probe arrangement

Procedure

1. Place the sample on the stage and lower the probes gently so all four **touch** it.
2. Connect the constant current source to the **outer** probes and the milli-voltmeter to the **inner** probes.
3. Set a current (e.g. 2 mA) and note the voltage V; vary the current in steps and note V each time.
4. Measure the **thickness w** of the sample (screw gauge) and the probe spacing s.
5. Calculate V/I and the resistivity rho for each reading; take the mean.
6. (Optional) Heat the sample in steps of 5 K, note V and I at each temperature, plot ln rho vs 1/T.

Observations

I (mA)	V (mV)	V/I (ohm)	rho (ohm m)
2			
4			
6			
8			

Calculations and graph

- rho = 4.532 w (V/I) for each row; **mean rho** = ... ohm m.
- A plot of **V against I** is a straight line through the origin; slope = V/I.

Result: Resistivity of the semiconductor rho = ... ohm m. (Band gap Eg = ... eV, if measured.)

Precautions

- Probes must make **good, gentle contact** (do not scratch the sample).

- Use a **small current** to avoid heating.
- Measure **w** accurately; it directly affects rho.

> * **Viva voce:**
> Q1. Why four probes? A: To avoid contact and lead resistance errors.
> Q2. How does rho of a semiconductor change with T? A: It decreases as T rises.
> Q3. Which probes carry current? A: The outer two.

Experiment 9 : Energy Gap using a p-n Junction Diode

Aim: To determine the **energy gap Eg of a semiconductor** using a **p-n junction diode** (reverse saturation current method).

Apparatus: p-n junction diode (Ge / Si), DC supply (about 3 V), micro-ammeter, thermometer, beaker with oil / water bath, heater (hot plate), connecting wires.

Principle / Theory

- In reverse bias the small current is the **reverse saturation current Is**, which depends strongly on temperature.
- **Is is proportional to exp(-Eg / kT)** (mainly), so log Is vs 1/T is a straight line.
- The **slope** of the line gives the energy gap.

> * **Formula:**
> log10(Is) = constant - Eg / (2.303 k T)
> Eg = 2.303 k x |slope| (J) or Eg (eV) = 1.984 x 10^-4 x |slope|
> k = 1.38 x 10^-23 J/K = 8.617 x 10^-5 eV/K

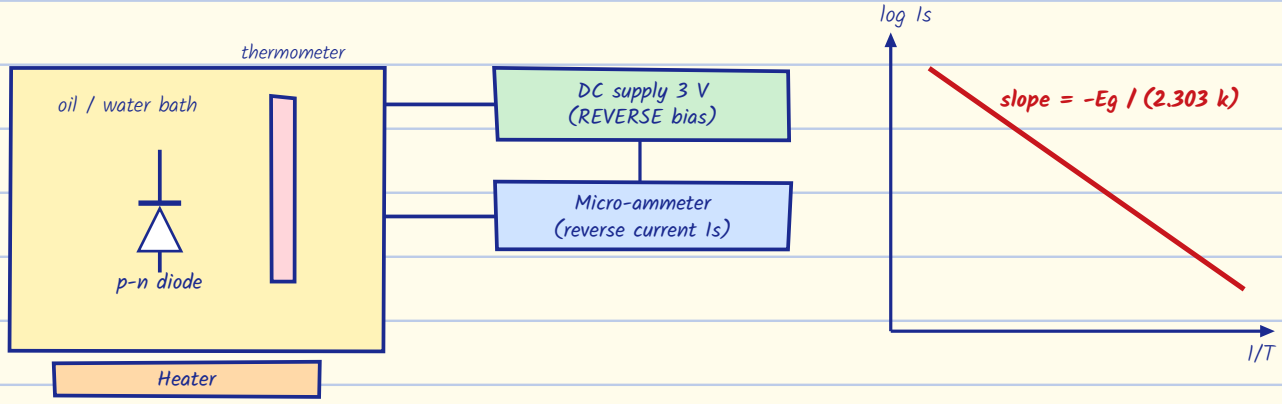


Fig 9.1 : Diode in a heated bath and log Is - 1/T graph

Procedure

1. Connect the diode in **reverse bias** with the micro-ammeter in series; immerse it in the bath with a thermometer.
2. Note the room temperature and the reverse current Is.
3. Heat the bath slowly; note Is at every **5 degree rise** up to about 80 deg C (stir the bath).
4. Convert temperature to kelvin: T = t + 273; find 1/T and log10 Is.
5. Plot **log10 Is (y) against 1/T (x)**: a straight line with negative slope.
6. Find the slope and calculate Eg.

Observations

t (deg C)	T (K)	1/T (x 10^-3 K^-1)	Is (micro-A)	log10 Is

Calculations and graph

- Slope = (y2 - y1) / (x2 - x1) from two far apart points (negative).
- Eg (eV) = 1.984 x 10^-4 x |slope|. Standard: **Ge = 0.67 eV, Si = 1.1 eV**.

Result: Energy gap of the semiconductor Eg = ... eV.

Precautions

- Never exceed the **maximum temperature** of the diode.
- Heat **slowly** and wait for **thermal equilibrium** before every reading.
- Keep the diode in **reverse bias** only.

> * **Viva voce:**
> Q1. What is the energy gap? A: Energy needed to take an electron from valence to conduction band.
> Q2. Why reverse bias? A: To get the temperature dependent saturation current.
> Q3. Eg of Ge and Si? A: 0.67 eV and 1.1 eV.

Experiment 10 : Circular Coil Field - Stewart & Gee

Aim: To study the **variation of the magnetic field along the axis of a circular coil** carrying current using a **Stewart and Gee** apparatus.

Apparatus: Stewart & Gee tangent galvanometer (circular coil on a stand with a deflection magnetometer / compass), battery, rheostat, ammeter, commutator (reversing key), plug key.

Principle / Theory

- The coil is kept in the **magnetic meridian**; the compass needle, free to rotate, sits on the **axis of the coil**.
- The coil field B (along the axis) is perpendicular to the **earth's horizontal field BH**, so the needle deflects by theta: **B = BH tan(theta)**.
- On the axis, B(x) = mu0 N I a^2 / [2 (a^2 + x^2)^(3/2)] - **maximum at the centre** and falling on both sides.
- The graph has **two points of inflexion**, separated by a distance equal to the **radius a** of the coil.

> * **Formula:**
> B = BH tan(theta) BH about 3.8 x 10^-5 T (use the local value)
> Theory: B = mu0 N I a^2 / [2 (a^2 + x^2)^(3/2)], mu0 = 4 pi x 10^-7 H/m

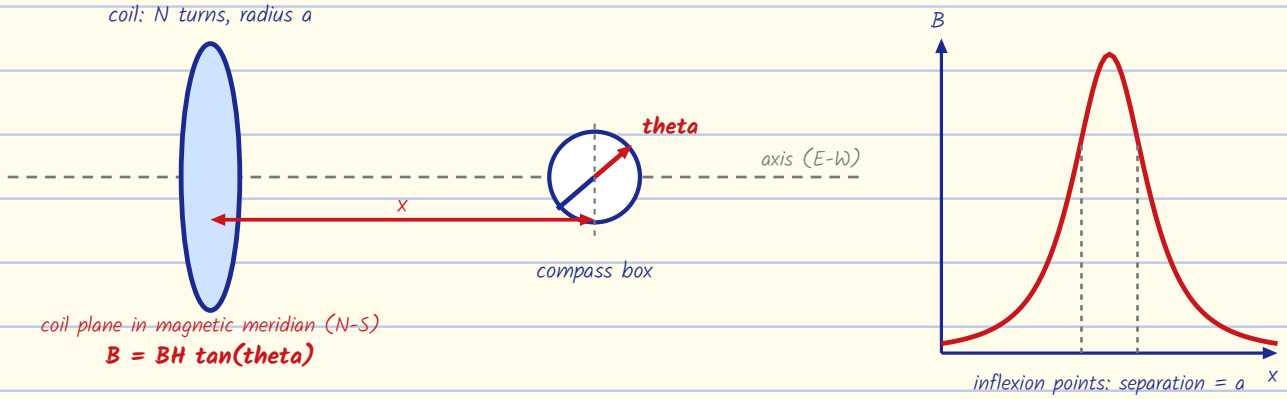


Fig 10.1 : Stewart and Gee set-up and B - x graph

Procedure

1. Set the coil in the **magnetic meridian** (needle reads 0-0 along the coil plane); check the compass box is on the axis.
2. Connect the battery, rheostat, ammeter and commutator in series with the coil.

3. Pass a steady current I (e.g. 1 A); note the deflections **theta1, theta2** at the coil centre.
4. Reverse the current and note the readings again; take the **mean theta**.
5. Move the compass along the axis in **steps of 2 cm** on one side and then the other; note theta each time.
6. Calculate B = BH tan(theta) and plot **B vs x**; mark the two inflexion points.

Observations

Distance x (cm)	theta1 (deg)	theta2 (deg)	Mean theta	tan(theta)	B (T)
0					
2					
4					
6					
8					
10					

Calculations and graph

- The distance between the **two points of inflexion** on the graph = radius **a** of the coil.
- Compare with the theoretical value of B at the centre = mu0 N I / (2a).

Result: The field is maximum at the centre and decreases along the axis; coil radius from the graph a = ... cm.

Precautions

- Remove **magnets and iron** objects from the table.
- Keep the plane of the coil in the **magnetic meridian**.
- Use the same current and **reverse it** to remove errors.

> * **Viva voce:**
> Q1. Shape of the B-x graph? A: Bell shaped, symmetric about the centre.
> Q2. What are inflexion points? A: Points where the curvature changes; at x = +-a/2.
> Q3. Why reverse the current? A: To cancel the error from the coil not being exactly in the meridian.

Experiment 11 : Hall Effect

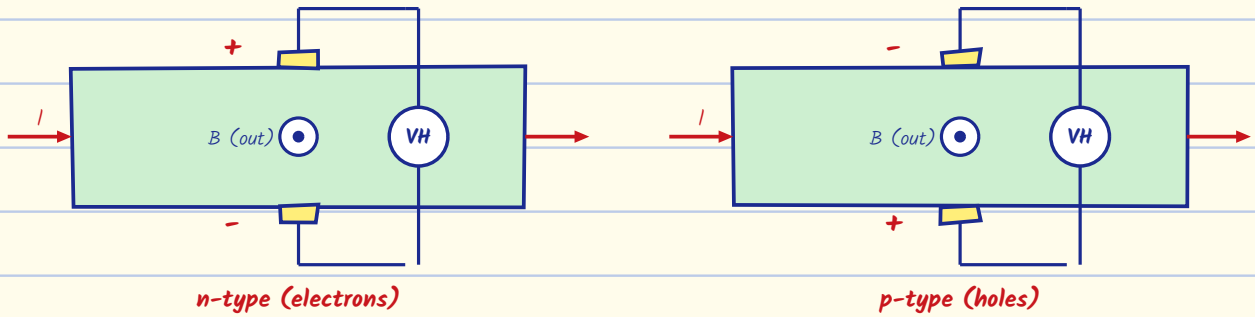
Aim: To determine the **Hall voltage and Hall coefficient** of a semiconductor and **identify its type (n or p)**.

Apparatus: Hall effect kit: semiconductor (Ge) crystal, electromagnet with constant current supply, constant current source, digital milli-voltmeter, Hall probe / gaussmeter.

Principle / Theory

- A current I flows along the crystal and a magnetic field B acts **perpendicular** to it; a transverse voltage **VH (Hall voltage)** appears.
- The sign of VH depends on the carriers: **n-type** (electrons) and **p-type** (holes) give **opposite polarity**.
- Hall coefficient **RH = VH t / (I B)**; carrier density **n = 1/(e |RH|)**; mobility = |RH| x conductivity.

> * **Formula:**
> VH = I B / (n e t) ; RH = VH t / (I B) (m^3/C)
> n = 1/(e |RH|) ; mobility mu = |RH| x sigma
> RH negative: n-type, RH positive: p-type



VH = I B / (n e t) -> the sign of VH tells the type

Fig 11.1 : Hall voltage in n-type and p-type samples

Procedure

1. Connect the crystal to the constant current source and the Hall voltage terminals to the milli-voltmeter.
2. With **B = 0**, note the offset voltage and adjust it to zero (or subtract it).
3. Fix the current (e.g. 5 mA); apply a magnetic field B and note **VH** for several B values.
4. Now keep B fixed and vary **I**; note VH again.

5. **Reverse** the magnetic field and the current and take the mean of VH to remove misalignment.

6. Plot **VH vs B** and **VH vs I** (straight lines); find RH. The **sign of VH** gives the type.

Observations

I (mA)	B (mT)	VH forward (mV)	VH reversed (mV)	Mean VH (mV)
5	100			
5	200			
5	300			
5	400			

Calculations and graph

- Slope of VH vs B at constant I: RH = (slope × t) / I.
- Calculate n = 1/(e |RH|) and state whether the sample is **n-type or p-type**.

Result: RH = ... m^3/C; carrier concentration n = ... m^-3; the sample is ..... -type.

Precautions

- Keep the sample in the **centre of the uniform field**.
- Do not exceed the **rated current** of the crystal.
- Use the same polarity convention throughout.

> * **Viva voce:**
> Q1. What is the Hall effect? A: A transverse voltage across a current carrying conductor in a perpendicular magnetic field.
> Q2. How is the type decided? A: From the sign of the Hall voltage.
> Q3. Unit of RH? A: m^3 per coulomb.

Experiment 12 : Temperature Coefficient of a Thermistor

Aim: To determine the **temperature coefficient of resistance** of a **thermistor**.

Apparatus: Thermistor (NTC), water / oil bath with heater, thermometer, digital multimeter (ohm range) or Wheatstone bridge, stirrer.

Principle / Theory

- A **thermistor** is a semiconductor resistor whose resistance changes strongly with temperature (NTC: **R decreases** as T rises).

- Its resistance follows **R = A e^(B/T)**, so **ln R vs 1/T is a straight line** with slope B (in kelvin).
- The temperature coefficient is **alpha = (1/R)(dR/dT) = -B / T^2** (negative for NTC).

> * **Formula:**
> ln R = ln A + B / T slope of (ln R vs 1/T) = B
> alpha = -B / T^2 (per kelvin)

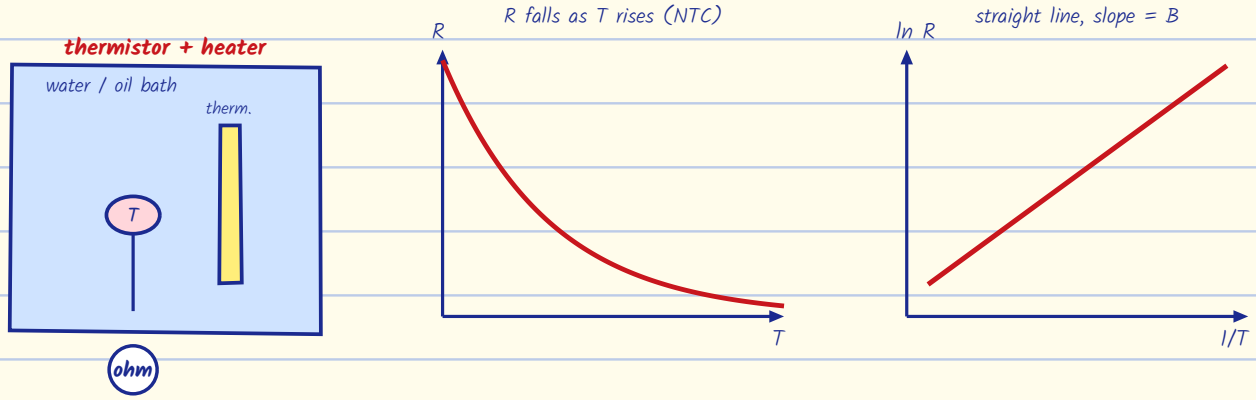


Fig 12.1 : Thermistor in a bath and its R - T behaviour

Procedure

1. Place the thermistor in a test tube in the bath with the thermometer; connect it to the multimeter.
2. Note the **room temperature** and the resistance R.
3. Heat the bath **slowly**; note R at every **5 deg C rise** up to about 80 deg C (stir).
4. Convert to kelvin; calculate 1/T and ln R.
5. Plot **ln R (y) vs 1/T (x)**; find the slope B.
6. Calculate alpha at a chosen temperature using alpha = -B / T^2.

Observations

t (deg C)	T (K)	1/T (x 10^-3)	R (ohm)	ln R

Calculations and graph

- Slope B = (ln R2 - ln R1) / (1/T2 - 1/T1) in kelvin.
- At T = 300 K: alpha = -B / 300^2 per kelvin (typical about -3 to -5 % per kelvin).

Result: Material constant B = ... K; temperature coefficient alpha = ... per kelvin at ..... K.

Precautions

- Heat **slowly** and stir for uniform temperature.
- Do not touch the thermistor leads to the **bath walls**.
- Avoid **self-heating**: use a small measuring current.

> * **Viva voce:**
> Q1. What is an NTC thermistor? A: Resistance falls as temperature rises.
> Q2. Why is alpha negative? A: Resistance decreases with temperature.
> Q3. Uses? A: Temperature sensors, inrush current limiters.

Experiment 13 : Compound Pendulum - g and Radius of Gyration

Aim: To determine the **acceleration due to gravity g** and the **radius of gyration k** using a compound pendulum.

Apparatus: Uniform bar with holes at equal spacing, knife-edge support, stop clock, metre scale.

Principle / Theory

- A rigid body swinging about a horizontal axis is a **compound pendulum**: T = 2 pi sqrt[(k^2 + h^2) / (g h)].
- h = distance from the pivot to the **centre of gravity G**; **k = radius of gyration about G**.
- The **T-d graph** has two symmetric branches; for a given T the four points A, B, C, D give **L = AC = BD** (equivalent simple pendulum).

> * **Formula:**
> T = 2 pi sqrt[(k^2 + h^2) / (g h)]
> g = 4 pi^2 L / T^2 k = sqrt(AG x BG) (from the graph)

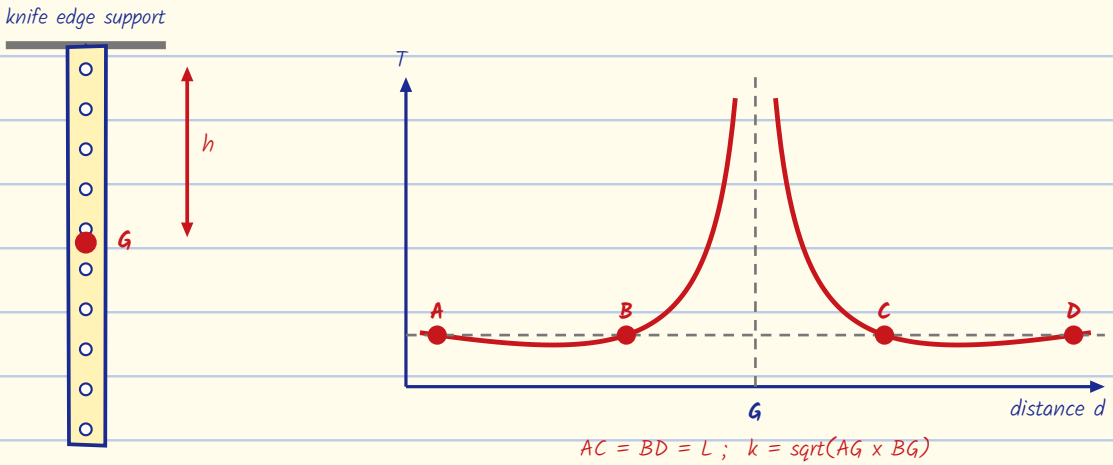


Fig 13.1 : Compound pendulum and the T - d graph

Procedure

1. Find the centre of gravity G of the bar (balance on a knife edge).
2. Suspend the bar through the **first hole**; give a small displacement (about 5 deg).
3. Note the time for **20 oscillations** and find T; repeat for every hole.
4. Plot **T (y) vs distance d (x)** from one end.
5. Draw a **horizontal line** cutting the curve at four points A, B, C, D; find L = AC = BD and k.
6. Calculate g = 4 pi^2 L / T^2 for several lines and take the mean.

Observations

Hole no.	Distance d (cm)	Time for 20 osc. t (s)	T = t/20 (s)
1			
2			
3			
4			
5			
6			
7			

Calculations and graph

- Mean g = ... m/s^2 (standard 9.8 m/s^2).
- Radius of gyration k from the graph; for a uniform rod k is about **length / sqrt(12)**.

Result: Acceleration due to gravity g = ... m/s^2; radius of gyration k = ... cm.

Precautions

- Use **small amplitudes** (about 5 deg).
- Start timing when the bar passes the **mean position**.
- Support the knife edge on a **rigid** table.

> * **Viva voce:**
> Q1. What is a compound pendulum? A: A rigid body free to oscillate about a horizontal axis.
> Q2. Where is T minimum? A: When h = k.
> Q3. What are conjugate points? A: Points of suspension giving the same T.

Experiment 14 : Magnetic Susceptibility - Kundt's Tube Method

Aim: To determine the **magnetic susceptibility of a paramagnetic liquid** by **Kundt's tube (Quincke's) method**.

Apparatus: Kundt's U-tube with one wide limb, FeCl3 or MnSO4 solution, electromagnet with variable DC supply, Hall probe / gaussmeter, travelling microscope, density bottle.

Principle / Theory

- A **paramagnetic liquid** is **drawn into** a strong magnetic field; a diamagnetic liquid is pushed out.
- The wide limb of the tube is placed **between the pole pieces**; when the field is on, the level **rises by h**.
- Magnetic pressure balances the liquid column: **(1/2) chi B^2 / mu0 = rho g h**.

> * **Formula:**
> chi = 2 mu0 rho g h / B^2 mu0 = 4 pi x 10^-7 H/m
> Mass susceptibility = chi / rho ; g = 9.8 m/s^2

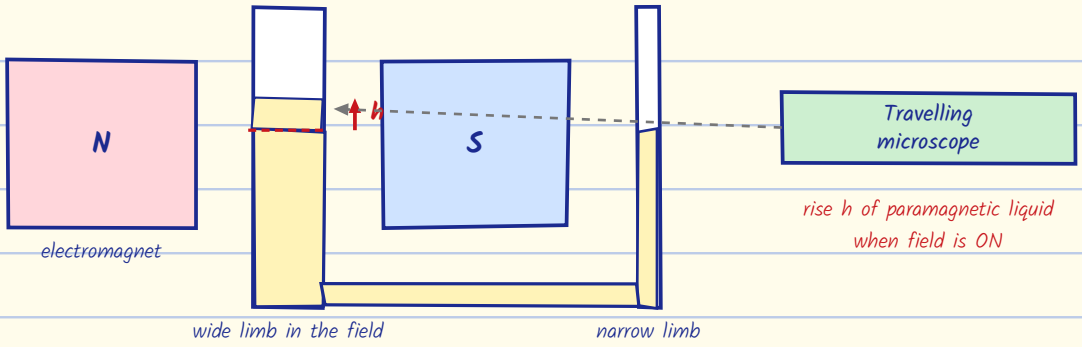


Fig 14.1 : Kundt's (Quincke's) tube between the poles

Procedure

1. Fill the U-tube with the solution; place the **meniscus of the wide limb** midway between the poles.
2. Focus the travelling microscope on the meniscus and note the **initial level** (field OFF).
3. Switch on the current (say 1 A); note the **new level** after the liquid settles; **h** = rise.
4. Measure the field B at the same position with the **gaussmeter / calibration graph** for each current.
5. Repeat for currents 1, 2, 3, 4 A; find the density rho with the density bottle.
6. Calculate chi for each B and take the mean; also plot **h vs B^2** (straight line).

Observations

Current (A)	B (T)	Initial level (cm)	Final level (cm)	h (cm)	chi
1					
2					
3					
4					

Calculations and graph

- chi = 2 mu0 rho g h / B^2 for each current; take the **mean chi**.
- A graph of **h vs B^2** is a straight line; slope = chi / (2 mu0 rho g).

Result: Magnetic susceptibility of the liquid chi = ... (positive, so the liquid is **paramagnetic**).

Precautions

- Keep the meniscus **exactly at the field centre**.
- Wait for the level to **settle** before noting.

- Keep the liquid **free of bubbles** and air.

> * **Viva voce:**
> Q1. What is susceptibility? A: Ratio of magnetisation to magnetising field, M/H.
> Q2. Sign for paramagnetic? A: Small positive.
> Q3. Why does the level rise? A: The liquid is attracted to the high field region.

Experiment 15 : Rigidity Modulus by Torsional Pendulum

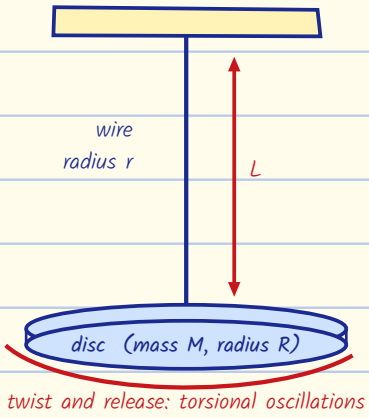
Aim: To determine the **rigidity modulus (eta) of the material of a wire** using a **torsional pendulum**.

Apparatus: Torsional pendulum (wire with a suspended disc), screw gauge, vernier calliper, metre scale, stop clock, physical balance.

Principle / Theory

- When the disc is twisted and released, it makes **torsional oscillations** with the wire acting as a spring.
- Torque per unit twist **C = pi eta r^4 / (2L)**; time period **T = 2 pi sqrt(I / C)**.
- For a disc, **I = M R^2 / 2**. Hence eta follows from T, L, r, M and R.

> * **Formula:**
> eta = 8 pi I L / (r^4 T^2) I = M R^2 / 2
> Typical steel: eta = 8 x 10^10 N/m^2



Restoring torque = C x angle

C = pi eta r^4 / (2L)

I = M R^2 / 2 (disc)

T = 2 pi sqrt(I / C)

Fig 15.1 : Torsional pendulum

Procedure

1. Measure the **mass M and radius R** of the disc; find I = M R^2 / 2.
2. Measure the **radius r of the wire** at several places with the screw gauge (two perpendicular directions).
3. Fix the **length L** of the wire between the clamp and the disc.
4. Give a small twist and release; note the time for **10 oscillations**; find T (three trials).
5. Change L (e.g. 40, 50, 60 cm) and repeat.
6. Calculate eta for each L and take the mean.

Observations

L (cm)	Time for 10 osc. t (s)	T (s)	T^2 (s^2)	eta (N/m^2)
40				
50				
60				
70				

Calculations and graph

- Wire radius r (mean) = ... mm (LC of screw gauge 0.01 mm).
- T^2 vs L is a **straight line**; slope = 4 pi^2 x 2 I / (pi eta r^4)... use eta = 8 pi I / (r^4 x slope).

Result: Rigidity modulus of the wire material eta = ... N/m^2.

Precautions

- Twist the disc **only by a small angle** (elastic limit).
- Avoid **swinging** of the disc; oscillations must be torsional only.
- Measure r accurately - it appears with **power 4**.

> * **Viva voce:**
> Q1. Define rigidity modulus. A: Ratio of shear stress to shear strain.
> Q2. Why is r measured carefully? A: eta depends on r^4.
> Q3. Time period depends on? A: I, L, r and eta.

Experiment 16 : Sonometer – Laws of a Stretched String

Aim: To **verify the laws of transverse vibrations of a stretched string** using a sonometer.

Apparatus: Sonometer with steel wire, set of tuning forks, weight hanger with slotted weights, movable bridge, paper rider, rubber pad.

Principle / Theory

- A stretched string of length l, tension T and linear density m vibrates with frequency **n = (1/2l) sqrt(T/m)**.
- **Law I:** n is proportional to 1/l (T, m constant). **Law II:** n is proportional to sqrt(T) (l, m constant). **Law III:** n is proportional to 1/sqrt(m) (l, T constant).
- Resonance with a tuning fork is detected when the **paper rider falls off**.

*** Formula:**

n = (1/2l) sqrt(T/m)

Law I: n x l = constant Law II: l / sqrt(T) = constant

Law III: l x sqrt(m) = constant T = M g

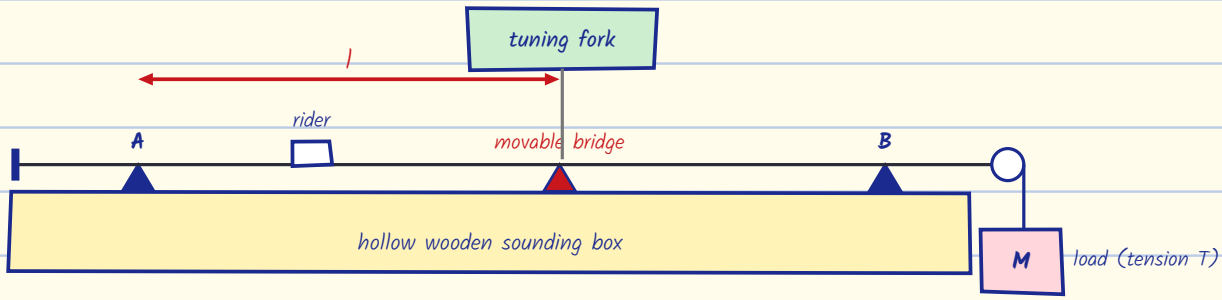


Fig 16.1 : Sonometer

Procedure

1. Fix a load M on the hanger (tension T = Mg) and place the paper rider on the wire.
2. Strike a tuning fork and press its stem on the box; move the **bridge** until the **rider is thrown off**; note the resonant length l.
3. **Law I:** repeat with forks of different frequencies n (T constant); check n x l.
4. **Law II:** use one fork and change the load M; find l for each; check l / sqrt(T).
5. Plot **l vs 1/n** (straight line) and **l^2 vs T** (straight line).

6. (Law III) use wires of different linear density at the same T and fork.

Observations

Fork n (Hz)	Load M (kg)	T = Mg (N)	l (cm)	n x l	l / sqrt(T)
256					
288					
320					
	1.0				
	1.5				
	2.0				

Calculations and graph

- Law I is verified if **n x l** is constant; Law II if **l / sqrt(T)** is constant.
- Find linear density m = mass/length of wire and check n = (1/2l) sqrt(T/m).

Result: The laws of transverse vibrations of a stretched string are verified. Linear density of wire m = ... kg/m.

Precautions

- The wire must be **uniform and free from kinks**.
- Use the **rubber pad** to hold the fork and press gently.
- Remove **friction** at the pulley.

> * **Viva voce:**
> Q1. Why does the rider fall off? A: Resonance makes the wire vibrate with large amplitude.
> Q2. How does frequency depend on length? A: n is inversely proportional to l.
> Q3. Which wire is used? A: Steel wire (high elasticity).

Experiment 17 : Young's Modulus by Non-uniform Bending

Aim: To determine **Young's modulus Y of the material of a wooden scale** by **non-uniform bending** (beam supported at two ends, loaded at the centre).

Apparatus: Wooden scale (metre scale), two knife edges, weight hanger with slotted weights, pin and travelling microscope (or optical lever), screw gauge, vernier calliper.

Principle / Theory

- The beam rests on two knife edges a distance **l** apart; a load **W = Mg** at the middle depresses the centre by **y**.
- From beam theory: **$y = W l^3 / (48 Y I_g)$** with **$I_g = b d^3 / 12$**.
- So Y is found from the slope of the **load – depression** line.

> * **Formula:**
> $Y = M g l^3 / (4 b d^3 y)$
> Graph: slope = M / y, so $Y = g l^3 \times \text{slope} / (4 b d^3)$

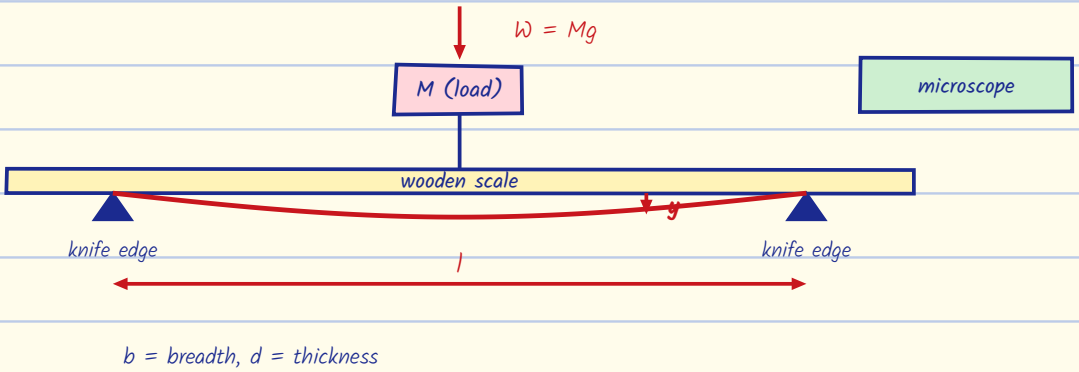


Fig 17.1 : Beam supported on two knife edges with a central load

Procedure

1. Measure the **breadth b and thickness d** of the scale at several places.
2. Place it on the two knife edges (length l apart) with the hanger at the **centre**.
3. Focus the microscope on the pin and note the reading with the **hanger only (zero load)**.
4. Add loads in steps (e.g. 50 g) and note the reading each time (**loading**); then remove loads (**unloading**).
5. Find the **mean reading** and the depression y for each load M.
6. Plot **M vs y** and calculate Y.

Observations

Load M (kg)	Loading (cm)	Unloading (cm)	Mean (cm)	Depression y (cm)
0.05				
0.1				
0.15				

Load M (kg)	Loading (cm)	Unloading (cm)	Mean (cm)	Depression y (cm)
0.2				
0.25				

Calculations and graph

- M/y from the graph; b = ... cm, d = ... cm, l = ... cm.
- Y = g l^3 x (M/y) / (4 b d^3) in N/m^2 (wood: about 10^10 N/m^2).

Result: Young's modulus of the wooden scale Y = ... N/m^2.

Precautions

- Place the load **exactly at the middle**.
- Keep the loads **within the elastic limit**.
- Measure d very accurately since Y depends on **d^3**.

> * **Viva voce:**
> Q1. Define Young's modulus. A: Ratio of longitudinal stress to longitudinal strain.
> Q2. What is non-uniform bending? A: Bending in which the curvature changes along the beam.
> Q3. Why is d measured carefully? A: Y depends on d cubed.

Experiment 18 : Frequency of a Tuning Fork – Melde's Experiment

Aim: To determine the **frequency of an electrically maintained tuning fork** by **Melde's experiment**.

Apparatus: Electrically maintained tuning fork, light thread, pulley, scale pan with weights, metre scale, physical balance.

Principle / Theory

- The fork vibrates the thread and a **stationary wave with loops** forms between the fork and the pulley.
- **Longitudinal arrangement** (prong along the string): string frequency = fork frequency. **Transverse arrangement** (prong perpendicular): string frequency = **half** of the fork frequency.
- Loop length = lambda/2; p loops in length L, tension T = (M + pan) g and linear density m.

> * **Formula:**
> Longitudinal: n = (p / 2L) sqrt(T/m)
> Transverse: n = (p / L) sqrt(T/m)
> T = (M + pan mass) g ; m = mass per unit length of thread

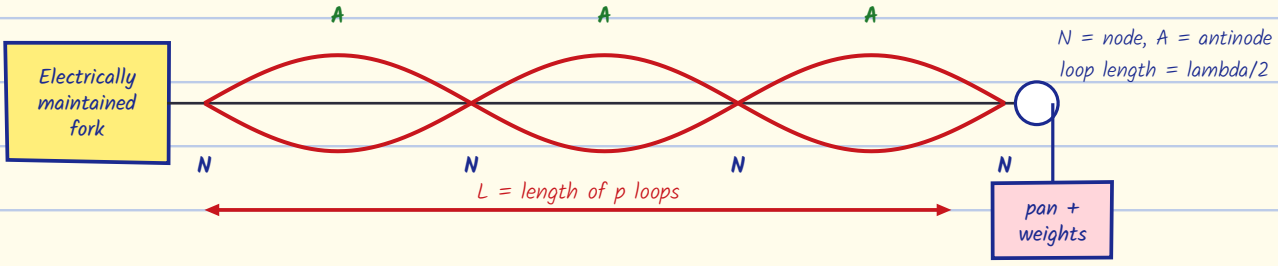


Fig 18.1 : Melde's experiment - stationary waves in a thread

Procedure

1. Find the **linear density m** of the thread (weigh 1 m of thread).
2. Attach one end of the thread to the prong, pass it over the pulley and hang the pan at the other end.
3. Start the fork; adjust the **weights** until sharp loops with clear nodes are formed.
4. Measure the **length L** of p loops (say 3 or 4) and note the total tension.
5. Repeat for different loads and for the **transverse** arrangement also.
6. Calculate n for each reading and take the mean.

Observations

Mode	M + pan (kg)	p	L (cm)	T (N)	n (Hz)
Longitudinal					
Longitudinal					
Transverse					
Transverse					

Calculations and graph

- Compute $n = (p / 2L)\sqrt{T/m}$ for longitudinal and $n = (p / L)\sqrt{T/m}$ for transverse; compare the results.
- $T p^2 / L^2$ should be nearly **constant** for a given mode.

Result: Frequency of the tuning fork n = ... Hz (longitudinal) and ..... Hz (transverse).

Precautions

- The thread must be **light and uniform**.
- Measure L from **node to node** (middle of loops), not from the fork end.

- Keep the **pulley frictionless**.

> * **Viva voce:**
> Q1. What are nodes and antinodes? A: Points of zero and maximum amplitude.
> Q2. Why half frequency in transverse mode? A: The string completes one vibration for two fork vibrations.
> Q3. Which wave is formed? A: Stationary (standing) transverse wave.

Formula Sheet : All 18 Experiments

Exp	Key formula	Result / constant
1 Newton's rings	R = (D_(n+p)^2 - D_n^2) / (4 p lambda)	R in cm (about 100)
2 Hg spectrum	lambda = sin(theta) / (n N)	Hg: 404.7 - 579 nm
3 Brewster	mu = tan(iB)	Glass: iB about 56 deg
4 Dielectric	C = I / (R |slope|); K = C / C0	K dimensionless
5 B-H curve	Loss = loop area (J/m^3 per cycle)	Br, Hc
6 Laser	lambda = d sin(theta) / n	He-Ne: 632.8 nm
7 Planck	h = e (V2 - V1) / (v2 - v1)	6.626 x 10^-34 J s
8 Four probe	rho = 4.532 w (V/I) (thin slab)	rho in ohm m
9 p-n diode	Eg = 2.303 k |slope| of log Is vs 1/T	Ge 0.67, Si 1.1 eV
10 Stewart-Gee	B = BH tan(theta)	BH about 3.8 x 10^-5 T
11 Hall effect	RH = VH t / (I B); n = 1/(e |RH|)	sign gives n / p type
12 Thermistor	alpha = -B / T^2	B from ln R vs 1/T
13 Compound pendulum	g = 4 pi^2 L / T^2	g = 9.8 m/s^2
14 Kundt's tube	chi = 2 mu0 rho g h / B^2	positive: paramagnetic
15 Torsion	eta = 8 pi I L / (r^4 T^2)	Steel: 8 x 10^10 N/m^2
16 Sonometer	n = (1/2l) sqrt(T/m)	n l = const
17 Young's modulus	Y = M g l^3 / (4 b d^3 y)	Wood: about 10^10 N/m^2
18 Melde	n = (p/2L) sqrt(T/m) ; (p/L) sqrt(T/m)	long. / trans.

> * **Exam tip:**
> In the lab exam write **Aim, Apparatus, Formula, Diagram, Procedure, Observation table, Calculation, Result and Precautions** in this order. A neat labelled diagram and graph earn the most marks.